WORKED SOLUTION

STEP I 1990, Question 16

Probability and Statistics • Triangular Distribution and Testing

STEP I 1990, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q16

The key idea

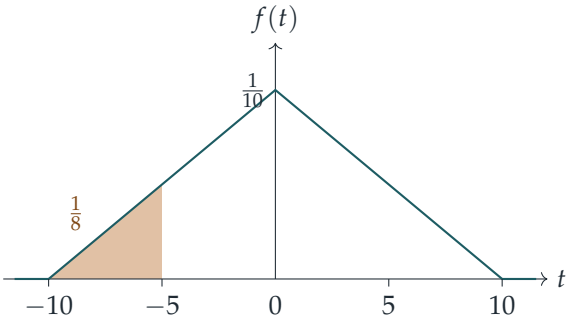
The density is a triangle, so every probability we need is the area of a small triangle, and the cumulative distribution function does all the work. The second thing to notice is that the 8 o'clock bus is the only bus that can turn up anywhere near 8 am: the 7 o'clock bus has gone by 7.10 and the 9 o'clock bus can't come before 8.50. So "catching a bus" in the morning really means "getting there before the 8 o'clock bus does". The final part then becomes a one-parameter binomial model and a significance test.

Solution

The density of T

The density rises linearly from 0 at $t = -10$ to a peak at $t = 0$, then falls at the same rate to 0 at $t = 10$, so it has the form $f(t) = k(10 - |t|)$ on $[-10, 10]$ and is zero elsewhere. The total area is that of a triangle with base 20 and height $10k$, so $\frac{1}{2} \cdot 20 \cdot 10k = 1$, giving $k = \frac{1}{100}$:

$$f(t) = \begin{cases} \dfrac{10+t}{100}, & -10 \le t \le 0, \\ \dfrac{10-t}{100}, & 0 \le t \le 10, \\ 0, & \text{otherwise.} \end{cases}$$



For later use, the cumulative distribution function on the left half is the area of a triangle with base $a+10$ and height $f(a)$:

$$\mathrm{P}(T<a)=\tfrac{1}{2}(a+10)\cdot\frac{a+10}{100}=\frac{(a+10)^2}{200},\qquad -10\le a\le 0,$$

and by symmetry about $t=0$, $\mathrm{P}(T>a)=\dfrac{(10-a)^2}{200}$ for $0\le a\le 10$. The shaded area in the figure is $\mathrm{P}(T<-5)=\frac{25}{200}=\frac{1}{8}$.

Catching a bus by 8.10 am

We assume that I wait at the stop until a bus comes. The bus due at 8 o'clock arrives at time T minutes after 8 am, with $-10\le T\le 10$; the previous bus has left by 7.10 and the next can't arrive before 8.50. So I catch a bus by 8.10 exactly when I reach the stop before the 8 o'clock bus arrives, and the 8.10 deadline is then automatically met.

If I reach the stop a minutes after 8 am, I catch it when $T>a$ (the chance of $T=a$ exactly is zero). The three cases give

$$\begin{aligned}
7{\cdot}55,\ a=-5:&\quad \mathrm{P}(T>-5)=1-\tfrac{25}{200}=\tfrac{7}{8},\\
7{\cdot}58,\ a=-2:&\quad \mathrm{P}(T>-2)=1-\tfrac{64}{200}=\tfrac{17}{25},\\
8{\cdot}02,\ a=2:&\quad \mathrm{P}(T>2)=\tfrac{64}{200}=\tfrac{8}{25}.
\end{aligned}$$

I reach the stop at 7.55 with probability 0.5, at 7.58 with probability $0.5\times 0.8=0.4$, and at 8.02 with probability $0.5\times 0.2=0.1$. My oversleeping has nothing to do with the bus, so by the law of total probability

$$\begin{aligned}
\mathrm{P}(\text{catch a bus by 8.10}) &= 0.5\cdot\tfrac{7}{8}+0.4\cdot\tfrac{17}{25}+0.1\cdot\tfrac{8}{25}\\
&= 0.4375+0.272+0.032=\frac{1483}{2000}=0.7415.
\end{aligned}$$

Is the new clock wrong?

Suppose first that the clock is correct, so I reach the stop at 7.55 every morning. Let T_8 and T_9 be the arrival times, in minutes after the hour, of the 8 o'clock and 9 o'clock buses; they are independent, each with density f. I am still waiting after 9 am exactly

when I miss the 8 o'clock bus ($T_8 < -5$) *and* the 9 o'clock bus is late ($T_9 > 0$). So on any one morning

$$p = \mathrm{P}(T_8 < -5)\,\mathrm{P}(T_9 > 0) = \tfrac{1}{8} \cdot \tfrac{1}{2} = \tfrac{1}{16}.$$

Ask yourself Which way should the test go: are 5 occasions "too many" or "too few"?

A wrongly set clock could push p either way. Say the clock is d minutes fast. Then I reach the stop at true time $7.55 - d$, and the clock shows 9.00 at true time $9.00 - d$, so the probability of waiting past "9 am" becomes

$$g(d) = \mathrm{P}(T_8 < -5 - d)\,\mathrm{P}(T_9 > -d).$$

A fast clock ($d > 0$) gets me there earlier, so I rarely miss the 8 o'clock bus and $g(d) < \frac{1}{16}$; a clock slightly slow does the opposite (for example $g(-2.5) \approx 0.079$). So the honest test is two-sided.

Let X be the number of the 100 mornings on which I wait past 9 am by the new clock. The arrival times of different buses are independent and I arrive at the same time every day, so the mornings are independent trials and

$$H_0:\ \text{clock correct}, \quad X \sim \mathrm{B}(100, \tfrac{1}{16}); \qquad H_1:\ \text{clock set wrongly, so } p \neq \tfrac{1}{16}.$$

Under H_0 we expect $\mathrm{E}(X) = 6.25$ occasions, with $\mathrm{Var}(X) = 100 \cdot \frac{1}{16} \cdot \frac{15}{16} \approx 5.86$, so the standard deviation is about 2.42. The observed 5 is just below the mean. Exactly,

$$\mathrm{P}(X \leq 5) = \sum_{k=0}^{5} \binom{100}{k} p^k (1-p)^{100-k} \approx 0.40 \qquad \text{with } p = \tfrac{1}{16},$$

and the normal approximation with continuity correction gives $\Phi((5.5 - 6.25)/2.42) = \Phi(-0.31) \approx 0.38$. This is nowhere near 0.025 (the lower tail of a two-sided 5% test), or indeed any usual significance level. So we do not reject H_0: **there is no evidence that the new clock is set incorrectly.**

That is not the same as evidence that the clock is right. In fact $g(d) = g(-5 - d)$, because the density is symmetric (swap the roles of the two buses), so a clock that is 5 minutes slow gives exactly the same $p = \frac{1}{16}$ as a correct one. No number of mornings of this kind of data could tell those two apart.

Checks

- The density has area 1, and the triangle formula agrees with numerical integration of f, for example $\mathrm{P}(T < -2) = 0.32$.
- The answer 0.7415 sits between the best case $\frac{7}{8}$ (always arriving at 7.55) and the worst case 0.32 (always at 8.02), as it should. A simulation of 10^6 mornings gave 0.741.
- A simulation of 10^6 pairs (T_8, T_9) gave 0.0622 for $\frac{1}{16} = 0.0625$, and the exact binomial tail $\mathrm{P}(X \leq 5)$ evaluates to 0.4002.

Common slips

- Forgetting that missing the 8 o'clock bus is necessary for waiting past 9: the answer $p = \frac{1}{2}$ (just "the 9 o'clock bus is late") ignores the first factor.
- Treating "catch a bus by 8.10" as needing $T \leq 10$ to be checked separately: the 8 o'clock bus always arrives by 8.10, so only $T > a$ matters.
- Running a one-tailed test for "too many" when the data show slightly too few; a wrong clock can move p in either direction.

Beyond STEP

The triangular density is exactly the distribution of the sum of two independent uniform random variables, a first example of a convolution, which is how densities of sums are found in a first-year probability course. The last part is a small taste of statistical identifiability: the symmetry $g(d) = g(-5 - d)$ means two different parameter values produce identical data, so no test can separate them. Grimmett and Welsh, *Probability: An Introduction*, covers convolutions and sums of random variables clearly.

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